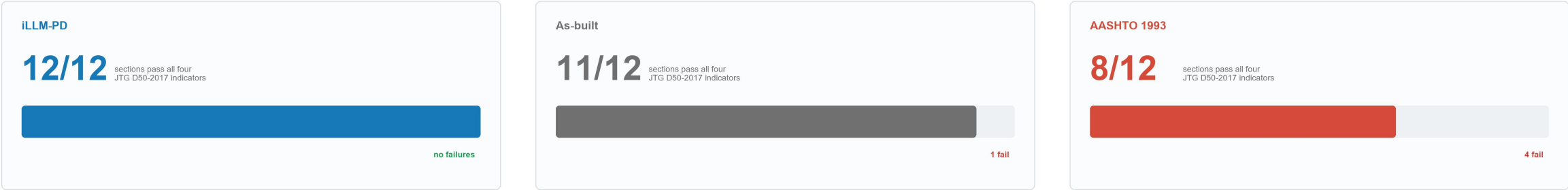


Fig. 4 | Comparison with established practice and cross-specification

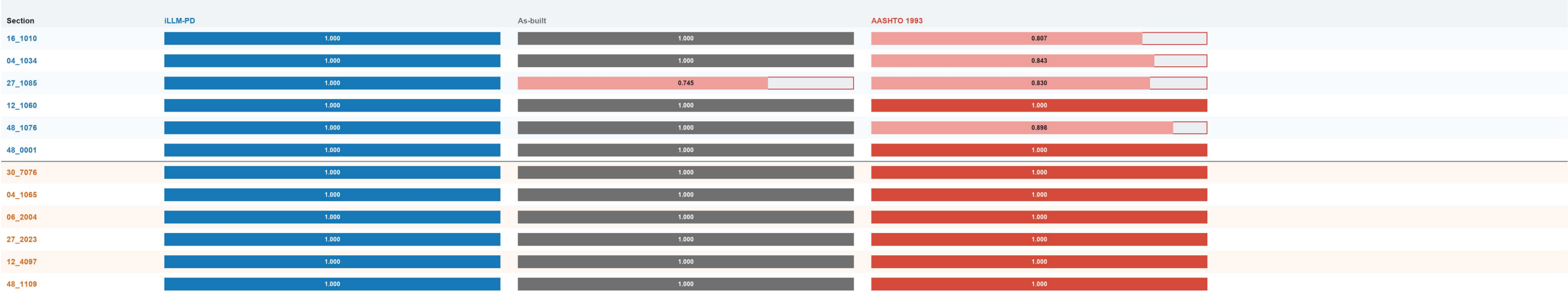
iLLM-PD is compared against as-built engineering practice, AASHTO 1993 designs, ME-PDG rutting predictions, and HPDS single-layer design.

a JTG compliance under identical criteria



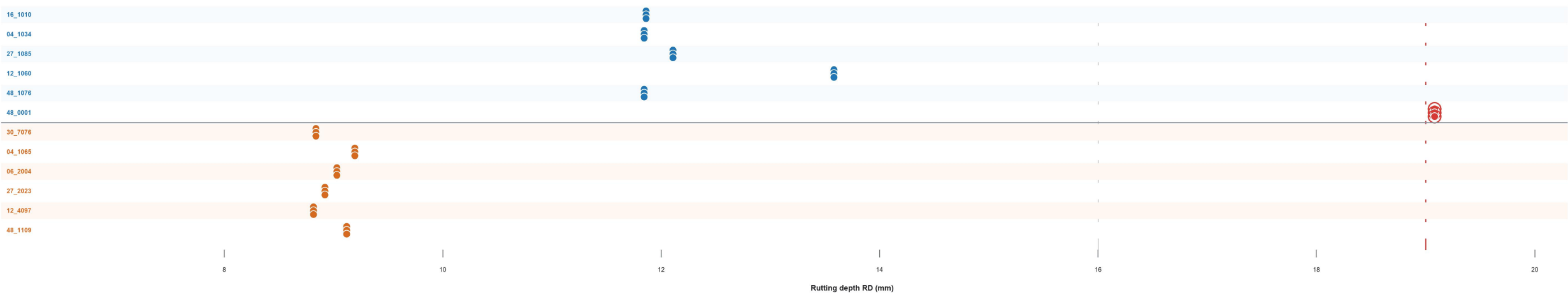
b Per-section design safety rate: failure locations

cell width is scaled to DSR; red edge marks DSR < 1.0

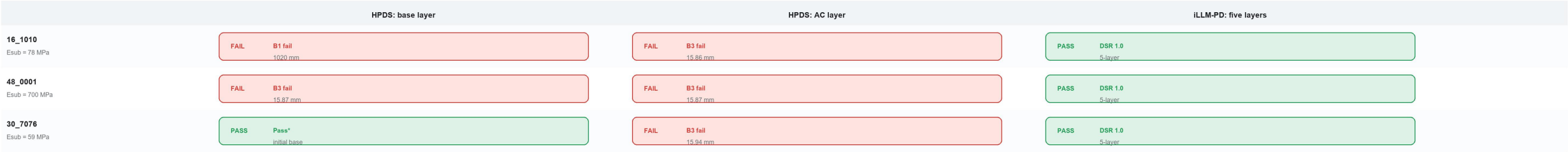


c ME-PDG rutting depth for 36 design-seed cases

33/36 pass; three failures are 48_0001 at 19.08 mm



d Paradigm contrast: HPDS single-layer adjustment vs iLLM-PD five-layer co-optimisation



* 30_7076: HPDS initial base thickness already satisfies the checks; this is verification, not iterative design.
HPDS adjusts one designated layer by construction, whereas iLLM-PD co-optimises all five layers.